A new normal-like inflated mean regression for bounded responses

Fernando Cortés-Tejada[a], Cristian L. Bayes[a,*], Luis Valdivieso[a]

[a] *Pontificia Universidad Católica del Perú*

Abstract

Bounded responses are often modelled by beta regressions, a distribution that does not perform well when fitting heavy tails or values near 0 or 1. If the boundaries have positive probability mass it leads to inflated models, which often model the inflated probabilities and the conditional mean $E[Y \mid Y \in (a, b)]$ separately, being a and b the boundaries. Sometimes the target of inference is the marginal mean and the data might have high density near 0 or 1 which leads these models to fail. In this article we propose three models based on the well-known normal distribution that allows to model data with high density near the boundaries and the marginal mean directly through covariates. There is no analytical solution to the proposed parametrization, therefore its solved numerically and the estimations are performed in the Bayesian mainframe. Simulation studies and application to a real medical case are included, both demonstrating a better fit of the proposed models.

Keywords: Tobit, Truncated-normal, Skew-normal, Inflated regression, Marginal mean model, Bayesian inference, MCMC

1. Introduction

Censored or bounded responses with positive mass probability in the boundaries appear in everyday life problems. For example, education grades in Perú

*Corresponding author

Email addresses: fcortes@pucp.pe (Fernando Cortés-Tejada), cbayes@pucp.edu.pe (Cristian L. Bayes), lvaldiv@pucp.edu.pe (Luis Valdivieso)

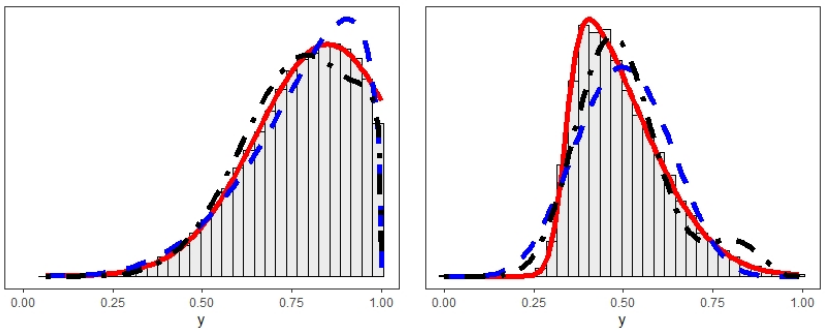
are bounded in a continuous range between 0 and 20, but can also be 0 or 20 with positive probability; the ratio of credit card line utilization is bounded to the interval [0,1] with positive probability in the boundaries (if someone does not use its credit card or uses its entire line). Medical research can also have bounded responses, such as reading accuracy for dyslexic/non-dyslexic children [1] given in the interval [0,1] or the Barthel index in NINDS RT-PA clinical trials [2] given in the interval [0, 100], both examples with positive probability in the boundaries.

A common distribution used to explain rates and proportions is the beta distribution allowing us to perform a beta regression [3], defined in the open interval (0,1), yet it can be rescaled to any open interval (a,b). The probability density function of the beta distribution tends to 0 when approaching the boundaries, unless it is a J-shaped, inverted-J-shaped, U-shaped or uniform, but in all the latter alternatives, it losses its bell-shape, a common shape when dealing with regressions. Even when it is bell-shaped, it is just symmetric when centered in $\frac{1}{2}$, if its mode lies in the interval $(0, \frac{1}{2})$ it necessarily becomes a right-skewed distribution and necessarily left-skewed for the mode in the interval $(\frac{1}{2}, 1)$, so it fails to fit translated symmetric responses or right-skewed distributions when translated to the right and vice versa.

Another problem of the beta distribution is that it fails to fit distributions with high density near the boundaries, which not necessarily represent heavy tails nor outliers. For example, in Figure 1a we compare the fit of a truncated normal distribution in (0,1) versus the Beta fit and the Flexible Beta fit [4] (the Beta rectangular [5] lead to mixture probability equal to 0, then is the same as the Beta fit). Even with 4 parameters, the state of the art Flexible Beta is not able to fully recover the shape of the distribution, yet, a simple normal distribution with 2 parameters can. Here can be clearly seen that the density lines of both beta distributions fall to 0 when $y \to 1$ leading to a lack of fit for values near 1. The same occurs when considering the symmetrical case.

The other case is when the distribution is left-skewed with mode in the interval $(0, \frac{1}{2})$ or right-skewed with mode in $(\frac{1}{2}, 1)$. For example, in Figure 1b

we compare the fit of a truncated skew-normal distribution in (0,1) versus the Beta fit and the Flexible Beta fit. The parameters of the skew-normal where chosen on purpose so that is right-skewed, left translated and that the best fit of the Beta gives a symmetrical bell-shaped distribution ($\alpha = \beta$), showing its lack of fit and ignoring the skewness of the distribution. The Flexible Beta recovers some of this skewness but still can not fit the distribution and induces a second mode near 0.8 which have no explanation for the data. The same occurs when considering the symmetrical case.



(a) Truncated normal (b) Truncated skew-normal

Figure 1: Figure (a) shows a histogram of 10,000 responses simulated from a zero-one truncated normal, with $\mu = 0.850$ and $\sigma = 0.200$. Red solid line is the real fit of the normal, blue dashed line is the fit of the Beta with $\alpha = 4.823$ and $\beta = 1.409$, black dot-dashed line is the fit of the Flexible Beta with $\alpha = 8.222, \beta = 1.332, \tau = 2.183$ and $p = 0.323$. Figure (b) shows a histogram of 10,000 responses simulated from a zero-one truncated skew-normal, with location $\mu = 0.338$, scale $\sigma = 0.200$ and skewness $\lambda = 6$. Red solid line is the real fit of the skew-normal, blue dashed line is the fit of the Beta with $\alpha = 7.333$ and $\beta = 7.333$, black dot-dashed line is the fit of the Flexible Beta with $\alpha = 11.239$, $\beta = 5.263$, $\tau = 7.283$ and $p = 0.106$. All these parameters are estimated from the maximum likelihood of each distribution.

When the boundaries a and b have positive mass density, the response is then a mixed random variable, with a discrete and a continuous component. Transformations of the response variable $y \in [a, b]$ into an $y^* \in (a, b)$ variable [6] have shown to induce bias in the estimation of true parameters [7] and also ignores the mixed nature of the variable. One way to solve this is extending the beta regression to a zero-one inflated beta regression [8; 9]. This extension can

be performed by considering any other distribution in the interval (a, b).

There are some cases where the target of inference is to model the probability of $Y = 0$ and the mean given the response is not zero $E[Y \,|\, Y \neq 0]$ separately, for example, in cases of count data, we can model the number of smoked cigarettes [10], and given that we are modelling each component separately, results can led to inferences between non-smokers ($Y = 0$) and the mean number of cigarettes of smokers $E[Y \mid Y \neq 0]$. In bounded cases, the credit card utilization ratio could be modelled, that would allow us model separately people that do not use its credit card and the ratio of people that uses it. This interest in modelling the effects separately does not mean that it is the only choice for understanding the problem.

On the other hand, when dealing with responses inflated in the upper boundary, it makes more sense to model the marginal mean instead, for example, if we have the GPA of the students in a school, it would be more interesting to model what factors (covariates) affect the mean performance of the school, instead of modelling separately the mean of the students that does not have the top GPA. This approach could favour making comparisons between schools and it sounds nicer for the school to say they want to take action to improve the marginal mean of the GPA instead of taking some students to the upper boundary GPA. Of course, this does not mean that we cannot model separately, for example, if the target of inference is understanding what characteristics favour a student to have the maximum GPA. Although some research have modelled the marginal mean in an inflated model [11; 7] or have approximated it [11; 12], it is still not a common practice.

As we have seen, there are countless cases in different subjects that involve bounded responses, like the ones mentioned before in medicine, banking and education. This makes important to have models that allow us to correctly fit these different cases and the ones that can appear in the future.

The literature is rapidly growing in presenting new modalities for fitting bounded responses , especially related to the Beta regression [3], including transformations from the closed interval to the open interval [6], solving problems

such as heavy tails and robustness in presence of outliers [13; 5; 14; 4] and cases of bimodality using identifiable mixtures [4].

The Beta rectangular and the Flexible Beta distributions (defined in (0,1)) are mixture models, the first is a mixture of a beta and a uniform distribution and the latter a mixture of two betas. The fact that they are mixtures often makes that the marginal mean is not the target of inference but the mean of each mixture component and in order to model the marginal mean, one needs to make a reparametrization. Even if we extend the model to be an inflated model, the latter marginal mean becomes the new conditional mean and needs to be reparametrized again, making the marginal mean of an inflated mixture model less intuitive to understand.

The beta rectangular is a mixture model with 3 parameters [13; 5], but it only allows to model heavy tails in both extremes of the domain, it fails when the data has just heavy tails in one side. On the other hand, the flexible beta [4] is a mixture of betas with 4 parameters, this distribution has more flexibility (thanks to the extra parameter) and solves the issues the beta rectangular has and is a robust for modelling with outliers and heavy tails in both or just one side. When we have an inflation in both boundaries, these models now have 5 and 6 parameters, respectively, thus they could be considered as complex models.

We can state that in the literature, models using the beta distribution have allowed solving problems related to modelling bounded responses and dealing with heavy tails and robustness with outlines. However, they have not yet been able to deal with the problems of high density near the boundaries (without being heavy tails nor outliers) neither with the inverse skewness depending on the position of the mode nor when the distribution is symmetrical and not centered in $\frac{1}{2}$. These problems will be solved by the models presented in this work.

In order to solve this problems and also modelling the marginal mean of inflated bounded responses directly through covariates, three models are proposed, the Tobit mean (Tobit-m), the Inflated truncated normal mean (ITNm)

and the Inflated truncated skew-normal mean (ITSNm), with 2, 4 and 5 parameters, respectively (considering already the inflation in both boundaries). Just by comparing the number of parameters we can see that these are simpler models that the ones using beta distribution. Moreover, the three distributions come from the best-known normal distribution, making it easier to comprehend. As of today, it has not yet been proposed a Zero-One inflated truncated normal-like model that allows to interpret the effects of covariates directly on the marginal mean, so we introduce it on the following sections.

Section 2 presents the three normal-like distributions, section 3 presents the model, 4 presents a simulation study, 5 presents an application to real data and 6 presents a discussion.

2. Distributions and regression models

In this section we present the 3 normal-like distributions that will be used, the reparametrization for modelling the marginal mean and the regression structure for each parameter.

The distributions presented here have been bounded to $[a, b]$, with $b > a$, but it is straightforward to transform them to the range $[0, 1]$ in order to be handled by the beta distribution.

2.1. Tobit

Tobit distribution [15], first studies [16], usually used for censored data [17] or bounded at a limited value [18].

The tobit distribution can be defined as a truncated normal in a and b, where the truncated densities accumulate as discrete positive mass probability in the boundaries. If $Y \sim Tobit_{ab}(\mu, \sigma)$, the its pdf can be written as:

$$f_Y(y \mid \mu, \sigma) = \begin{cases} \Phi(a \mid \mu, \sigma), & \text{if } Y \leq a \\ 1 - \Phi(b \mid \mu, \sigma), & \text{if } Y \geq b \\ \phi(y \mid \mu, \sigma), & \text{if } Y \in (a, b) \end{cases} \tag{1}$$

where $\phi(\cdot \mid \mu, \sigma)$ and $\Phi(\cdot \mid \mu, \sigma)$ are the pdf and cdf of the normal distribution with mean μ and standard deviation σ, respectively.

The mean of the tobit distribution is:

$$\begin{aligned} E[Y] &= a\,\Phi(\alpha) + b\,[1-\Phi(\beta)] + [\Phi(\beta)-\Phi(\alpha)]\;E[Y \mid Y \in (a,b)] \\ &= a\,\Phi(\alpha) + b\,[1-\Phi(\beta)] + \mu\,[\Phi(\beta)-\Phi(\alpha)] - \sigma\,[\phi(\beta)-\phi(\alpha)] \end{aligned} \tag{2}$$

where $\alpha = \frac{a-\mu}{\sigma}$ and $\beta = \frac{b-\mu}{\sigma}$.

2.2. Inflated truncated normal

Truncated normal [19].

The inflated truncated normal (ITN) distribution, $Y \sim ITN_{ab}(\delta_a, \delta_b, \mu, \sigma)$, is a generalization of the Tobit, obtained by increasing the number of parameters from 2 to 4. Its pdf can be written as:

$$f_Y(y \mid \delta_a, \delta_b, \mu, \sigma) = \begin{cases} \delta_a, & \text{if } Y \le a \\ \delta_b, & \text{if } Y \ge b \\ \delta_{ab}\,\frac{\phi(y|\mu,\sigma)}{\Phi(b|\mu,\sigma)-\Phi(a|\mu,\sigma)}, & \text{if } Y \in (a,b) \end{cases} \tag{3}$$

where $\delta_{ab} = 1 - \delta_a - \delta_b$. Look that if $\delta_a = \Phi(a \mid \mu, \sigma)$ and $\delta_b = 1 - \Phi(b \mid \mu, \sigma)$ we will return to the special case (1).

The mean of the inflated truncated normal is:

$$\begin{aligned} E[Y] &= a\,\delta_a + b\,\delta_b + (1-\delta_a-\delta_b)\,E[Y|Y \in (a,b)] \\ &= a\,\delta_a + b\,\delta_b + (1-\delta_a-\delta_b)\left(\mu - \sigma\,\frac{\phi(\beta)-\phi(\alpha)}{\Phi(\beta)-\Phi(\alpha)}\right) \end{aligned} \tag{4}$$

where $\alpha = \frac{a-\mu}{\sigma}$ and $\beta = \frac{b-\mu}{\sigma}$.

2.3. Inflated truncated skew-normal

Skew-normal distribution [20], Skew-normal approximation [21], Truncated skew-normal [22].

Bayesian inference for the skewness parameter [23]

Following the path, the inflated truncated skew-normal (ITSN), $Y \sim ITSN_{ab}(\delta_a, \delta_b, \mu, \sigma, \lambda)$ is a generalization of the ITN, obtained by adding a skewness parameter λ. Its pdf can be written as:

$$f_Y(y \mid \delta_0, \delta_1, \mu, \sigma, \lambda) = \begin{cases} \delta_a, & \text{if } Y \leq a \\ \delta_b, & \text{if } Y \geq b \\ \delta_{ab} \, \frac{\frac{2}{\sigma}\phi(\frac{y-\mu}{\sigma})\Phi(\lambda(\frac{y-\mu}{\sigma}))}{F(b|\mu,\sigma,\lambda)-F(a|\mu,\sigma,\lambda)}, & \text{if } Y \in (a,b) \end{cases} \tag{5}$$

where $\delta_{ab} = 1 - \delta_a - \delta_b$ and $F(\cdot \mid \mu, \sigma, \lambda)$ is the cdf of the skew-normal with location parameter μ, scale parameter σ and skewness parameter λ. Look that if $\lambda = 0$ then we will return to the special case (3).

The mean of the inflated truncated skew-normal is:

$$\begin{aligned} E[Y] &= a\,\delta_a + b\,\delta_b + (1 - \delta_a - \delta_b)\, E\left[Y \mid Y \in (a,b)\right] \\ &= a\,\delta_a + b\,\delta_b + (1 - \delta_a - \delta_b) \left(\mu - \sigma \, \frac{X + \frac{2}{\sqrt{2\pi}} \frac{\lambda}{\lambda_s} \cdot W}{Z} \right) \end{aligned} \tag{6}$$

where $\lambda_s = \sqrt{1 + \lambda^2}$,

$$\begin{aligned} X &= 2\phi(\beta)\Phi(\lambda\beta) - 2\phi(\alpha)\Phi(\lambda\alpha), \\ W &= \Phi(\lambda_s\beta) - \Phi(\lambda_s\alpha), \\ Z &= F_\lambda(\beta) - F_\lambda(\alpha), \end{aligned}$$

and $F_\lambda(\cdot)$ is the cdf of the standard skew-normal with skewness parameter λ, $\alpha = \frac{a-\mu}{\sigma}$ and $\beta = \frac{b-\mu}{\sigma}$.

2.4. Reparametrization for modelling the marginal mean

The marginal mean of (1), (3) and (5) has the general form

$$E[Y] = a\,\delta_a + b\,\delta_b + (1 - \delta_a - \delta_b)\, E\left[Y \mid Y \in (a,b)\right] \tag{7}$$

where $E\left[Y \mid Y \in (a,b)\right]$ varies between the 3 distributions and δ_a and δ_b have fixed values in the Tobit distribution.

In order to perform a regression over the marginal mean, we introduce γ, such that:

$$\gamma = a\,\delta_a + b\,\delta_b + (1 - \delta_a - \delta_b)\,E\left[Y \mid Y \in (a, b)\right], \tag{8}$$

however, γ has the constraint $a + \delta_b(b-a) < \gamma < b - \delta_a(b-a)$, therefore we introduce

$$\alpha_a = \frac{(b-a)\delta_a}{b-\gamma} \quad \text{and} \quad \alpha_b = \frac{(b-a)\delta_b}{\gamma - a}, \tag{9}$$

so the constraints are broken, the marginal mean γ can vary in (a, b) and the parameters related to the inflation α_a and α_b can vary in (0,1). Both σ and λ are left the same. From now on, the distributions (1), (3) and (5) under parametrization (8) and (9) will be referred as Tobit-m, ITNm and ITSNm, respectively, where m stands for mean parametrization.

The backwards reparametrization for $\delta_a, \delta_b, \sigma$ and λ is straightforward, but not for μ. It can not be solved analytically from (8) due to its complexity. Two approaches can be used to solve this problem: the first consists in solving numerically the equation for μ with $\gamma, \alpha_a, \alpha_b, \sigma$ and λ given; the second consists in approximating γ as an easier or well-known function of μ, so the inverse can be computed with $\gamma, \alpha_a, \alpha_b, \sigma$ and λ given. A discussion and implementation of these two approaches can can be found in Section 3.

2.5. Regression models

In order to make the regressions easier, the response variable will be re-scaled to the interval (0,1) without loss of generality. The proposed regression will be performed on the parameters of the ITSNm since its the most general model, but it is straightforward to simplify it for the Tobit-m and the ITNm. The regressions are the following:

$$g_1(\gamma_i) = \boldsymbol{x}_{1i}^{\intercal}\boldsymbol{\beta_1}$$
$$g_2(\alpha_{0i}) = \boldsymbol{x}_{2i}^{\intercal}\boldsymbol{\beta_2}$$
$$g_3(\alpha_{1i}) = \boldsymbol{x}_{3i}^{\intercal}\boldsymbol{\beta_3}$$
$$g_4(\sigma_i) = \boldsymbol{x}_{4i}^{\intercal}\boldsymbol{\beta_4}$$
$$g_5(\lambda_i) = \boldsymbol{x}_{5i}^{\intercal}\boldsymbol{\beta_5}$$

The link functions g_1, g_2 and g_3 are strictly monotone functions from (0,1) to $\mathbb{R}$ (e.g. logit function). The link function g_4 is a strictly monotone function from $\mathbb{R}^+$ to $\mathbb{R}$ (e.g. log function). The link function g_5 is a strictly monotone function from $\mathbb{R}$ to $\mathbb{R}$ (e.g. identity function).

3. The model

In this section we present the likelihood, bayesian inference for the posteriori, the sampling algorithms and the model comparison criteria

3.1. Likelihood

Let $\boldsymbol{Y} = [Y_1, Y_2, ..., Y_n]^{\intercal}$, be n independent mixed bounded responses vector and the parameter vector $\boldsymbol{\theta} = [\boldsymbol{\beta_1}, \boldsymbol{\beta_2}, \boldsymbol{\beta_3}, \boldsymbol{\beta_4}, \boldsymbol{\beta_5}]^{\intercal}$. The likelihood conditional on the data is:

$$L(\boldsymbol{\theta} \mid \boldsymbol{Y}) = \prod_{i=1}^{n} \delta_{0i}^{I(y_i=0)} \delta_{1i}^{I(y_i=1)} \left[(1-\delta_{0i}-\delta_{1i}) f_Y(y_i \mid \mu_i, \sigma_i, \lambda_i)\right]^{I(y_i \in (0,1))} \quad (10)$$

where $I(.)$ is the indicator function and f_Y is the truncated skew-normal pdf, as the general case, but it must be seen that we can easily get to the other distributions likelihood by setting the parameters appropriately. The regressions

proposed for this model are:

$$\begin{aligned} logit(\gamma_i) &= \boldsymbol{x}_{1i}^{\intercal}\boldsymbol{\beta_1} \\ logit(\alpha_{0i}) &= \boldsymbol{x}_{2i}^{\intercal}\boldsymbol{\beta_2} \\ logit(\alpha_{1i}) &= \boldsymbol{x}_{3i}^{\intercal}\boldsymbol{\beta_3} \\ log(\sigma_i) &= \boldsymbol{x}_{4i}^{\intercal}\boldsymbol{\beta_4} \\ \lambda_i &= \boldsymbol{x}_{5i}^{\intercal}\boldsymbol{\beta_5}. \end{aligned} \tag{11}$$

Note that (10) is not expressed in directly in terms of (11) directly. However, this values can be computed from:

$$\begin{aligned} \alpha_0 &= \frac{\delta_0}{1-\gamma} \\ \alpha_1 &= \frac{\delta_1}{\gamma} \\ \sigma &= \sigma \\ \lambda &= \lambda \\ \mu &= f^{-1}(0) \end{aligned} \tag{12}$$

where $f(\mu) = \delta_0 a + \delta_1 b + (1 - \delta_0 - \delta_1)\, E\left[Y \mid Y \in (a, b)\right] - \gamma$.

3.2. Posterior

All parameters are assumed to be independent, thus their priori is

$$P(\boldsymbol{\theta}) = P(\boldsymbol{\beta_1})P(\boldsymbol{\beta_2})P(\boldsymbol{\beta_3})P(\boldsymbol{\beta_4})P(\boldsymbol{\beta_5})$$

where each priori distribution is set to a multivariate normal, centered in 0 with large diagonal covariance matrix in order to make it uninformative. Finally the parameters will be sampled from

$$P(\boldsymbol{\theta} \mid \boldsymbol{y}) \propto L(\boldsymbol{\theta} \mid \boldsymbol{y}) \times P(\boldsymbol{\theta}). \tag{13}$$

3.3. Sampling algorithms

In order to sample from (13), the first approach of solving μ numerically is introduced in Algorithm 1. It should be noted that this algorithm is proposed for the specific case of ITNm with constant parameters bounded in (a,b), but its generalization for other distributions, regressions and boundaries is straightforward by using the equations previously stated.

The advantages of Algorithm 1 are that root-solving algorithms are now implemented in every programming language and there are packages that allow the user to control the norm of the error, the relative and absolute tolerance, the number of iterations and other parameters [24], which make estimations of the root as precise as we want. However, for Algorithm 1, a root has to be solved M times, where M is the total number of iterations, and if we consider a regression of n subjects, then each one has a different μ, therefore $M \times n$ roots have to be solved. Since the function depends on at least two normal pdf and two normal cdf, it is not really fast for the algorithms to compute the root, so scalability is a problem for this algorithm, especially when the number of iterations M has to be large in order to converge or when a large dataset is fitted.

The second approach consist on finding an approximated function of γ in terms of μ so μ could be computed from the inverse of that function. The first thoughts are always to approach the most complex part first, in this case ϕ and Φ. There are several proposed approximations for the normal pdf [25] and for the normal cdf [26] and more to come, however is this case they do not help simplifying the problem, on the contrary, some of the approximations in the literature depend on integrals or are piecewise-defined functions [21; 27], thus this path was not taken. Instead of trying to approximate each function separately let us consider the tobit mean γ with boundaries in 0 and 1 (the most common case):

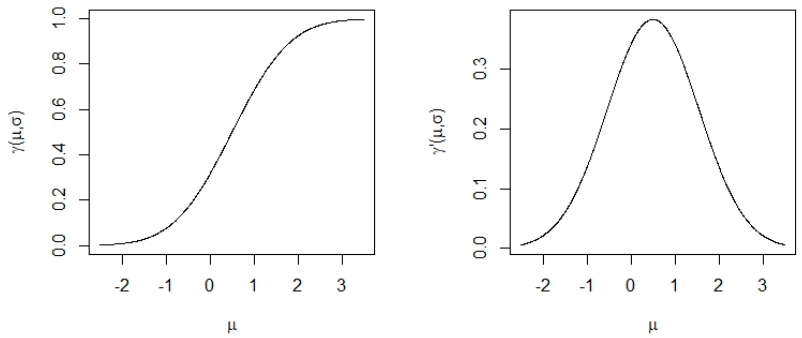
$$\gamma(\mu, \sigma) = 1 - \Phi(\beta) + \mu \left[\Phi(\beta) - \Phi(\alpha)\right] - \sigma \left[\phi(\beta) - \phi(\alpha)\right] \tag{14}$$

where $\alpha = \frac{-\mu}{\sigma}$ and $\beta = \frac{1-\mu}{\sigma}$. For a fixed σ (i.e. $\sigma = 1$) equation (14) has a

sigmoid shape, similar to a normal cdf as can be seen in Figure 2a. Then, one can perform the partial derivative with respect to μ, that we will call γ':

$$\frac{\partial \gamma(\mu, \sigma)}{\partial \mu} = \gamma'(\mu, \sigma) = \Phi(\beta) - \Phi(\alpha) \tag{15}$$

and we can see its plot on Figure 2b, which is similar to a normal pdf. So the idea is to proof (15) is a pdf, approximate this pdf by a known distribution pdf and given γ and σ estimate μ via the quantile function.



(a) γ as a cdf

(b) γ' as a pdf

Figure 2: Figure (a) shows γ as a function of μ with $\sigma = 1$, which looks like a normal cdf. Figure (b) shows γ' as a function of μ with $\sigma = 1$, which looks like a normal pdf.

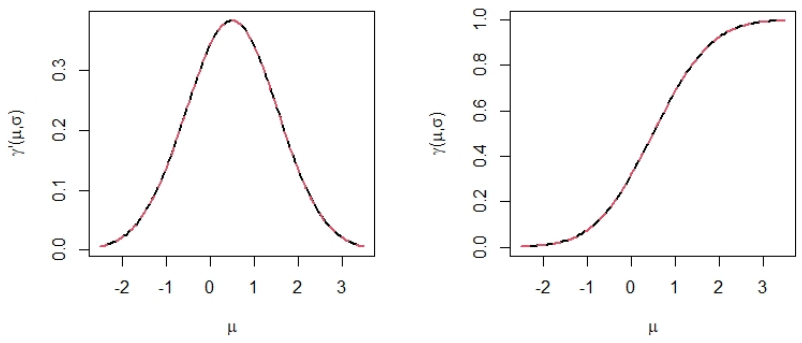
To prove γ' is a pdf we use the integrals in [28]:

$$\int_{-\infty}^{\infty} \gamma' d\mu = \int_{-\infty}^{\infty} (\Phi(\beta) - \Phi(\alpha))\, d\mu = 1. \tag{16}$$

Therefore, we can use the method of moments to find the two parameters (μ, σ) of the normal distribution we will use to approximate γ'. The first two moments of (15) are:

$$m_1 = \frac{1}{2} \text{ and } m_2 = \frac{1}{3} + \sigma^2, \tag{17}$$

so for $\sigma = 1$ we have the approximated $\gamma'(\mu, 1)$ and $\gamma(\mu, 1)$ in Figures 3a and 3b, respectively.



(a) γ' and its approximation

(b) γ and its approximation

Figure 3: Figure (a) shows γ' as a function of μ with $\sigma = 1$ in black, and shows its approximation by a normal pdf in red dashed lines. Figure (b) shows γ as a function of μ with $\sigma = 1$ in black, and shows its approximation by a normal cdf in red dashed lines.

The approximation works well for $\sigma \geq 1$ but when σ gets smaller this approximation fails to fit the real curve as can be seen in the red dashed lines in Figure 4a and Figure 4b.

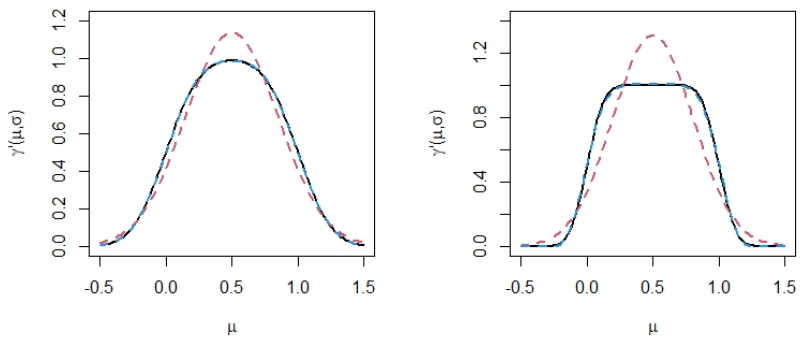
In order overcome this lack of fit by a normal curve and to approximate this new shape we propose to use the exponential power distribution and we will need to add the 3rd and 4th moment of γ':

$$m_3 = \frac{1}{4} + \frac{3}{2}\sigma^2 \text{ and } m_4 = \frac{1}{5} + 2\sigma^2 + 3\sigma^4, \tag{18}$$

the K moment of γ' can be generalized but is left for future work. (se podrían incluir las integrales de los 4 primeros momentos como apéndice e incluso encontrar la fórmula recurrente para los momentos, se podría plantear hacerlos para los truncamientos a y b generales en lugar de 0 y 1)

Let X be a random variable that follows an exponential power distribution, $X \sim EP(\mu, \sigma, p)$ then its pdf is given by:

$$f_X(x \mid \mu_{ep}, \sigma_{ep}, p) = \frac{p}{2\Gamma(\frac{1}{p})\sigma_{ep}} exp\left[-\left(\frac{|x - \mu_{ep}|}{\sigma_{ep}}\right)^p\right], \tag{19}$$



(a) γ' with $\sigma = 0.2$ (b) γ' with $\sigma = 0.1$

Figure 4: Figure (a) shows γ' as a function of μ with $\sigma = 0.2$ in black, and shows its approximations by a normal pdf in red dashed lines and by an exponential power pdf in blue dashed lines. Figure (b) shows γ as a function of μ with $\sigma = 0.1$ in black, and shows its approximations by a normal pdf in red dashed lines and by an exponential power pdf in blue dashed lines.

where we use μ_{ep} and σ_{ep} to differentiate the exponential power parameters from γ' parameters. Figure 5a shows some shapes of the exponential power distribution, where it should be noted that when $p = 1$ it is the Laplace distribution and when $p = 2$ is the normal distribution. The first four moments of X are given by [29]:

$$\begin{aligned} E(X) &= \mu_{ep}, \\ E(X^2) &= \mu_{ep}^2 + \frac{\sigma_{ep}^2 \Gamma(\frac{3}{p})}{\Gamma(\frac{1}{p})}, \\ E(X^3) &= \mu_{ep}^3 + \frac{3\mu_{ep}\sigma_{ep}^2 \Gamma(\frac{3}{p})}{\Gamma(\frac{1}{p})} \quad \text{and} \\ E(X^4) &= \mu_{ep}^4 + \frac{6\mu_{ep}^2\sigma_{ep}^2 \Gamma(\frac{3}{p})}{\Gamma(\frac{1}{p})} + \frac{\sigma_{ep}^4 \Gamma(\frac{5}{p})}{\Gamma(\frac{1}{p})}. \end{aligned} \tag{20}$$

The exponential power distribution has 3 parameters, however, since it is symmetrical, the 3rd moment gives no information and so the 4th needs to be used. Using the moments of γ' (17) and (18) and matching the exponential

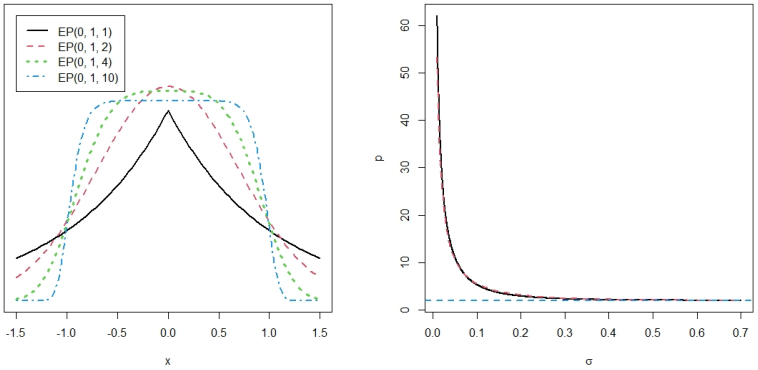
power moments (20) we obtain the following relations:

$$\mu_{ep} = \frac{1}{2}, \tag{21a}$$

$$\frac{\Gamma(\frac{1}{p})\Gamma(\frac{5}{p})}{\Gamma^2(\frac{3}{p})} = \frac{m_4(\sigma) - \frac{3}{2}m_2(\sigma) + \frac{3}{8}}{(m_2(\sigma) - \frac{1}{4})^2} = M(\sigma), \tag{21b}$$

$$\sigma_{ep} = \sqrt{\left(m_2(\sigma) - \frac{1}{4}\right)\frac{\Gamma(\frac{1}{p})}{\Gamma(\frac{3}{p})}} \tag{21c}$$

where we can see from equation (21b) that the power parameter p depends only on the given σ of γ', thus if we know σ we know p and therefore σ_{ep} from equation (21c). We can see the relation between σ and p in Figure 5b, where it can be noticed that for values of $\sigma > 0.6$ (insertar el valor exacto cuando sigma es 0.6), the value of $p \approx 2$ so a normal pdf can be used.



(a) Exponential power pdf

(b) p as a function of σ

Figure 5: Figure (a) shows the exponential power pdf for different values of p. Figure (b) shows p as a function of σ in black with its asymptotic behaviour approaching to 2 as σ increases, and an approximation by $2coth(\frac{15}{4})$ in a red dashed line.

The approximations of γ' by the exponential power using the method of moments can be seen in the blue dashed lines in Figures 4a and 4b, for $\sigma = 0.2$ and $\sigma = 0.1$ respectively. Given this good approximation of γ' it is straightforward that γ will be approximated by the cdf of the exponential power, and that given

γ and σ, we can compute the quantile function in order to obtain the approximation of μ. This can be easily implemeted in R with the package *gamlss.dist* [30] which contains the pdf, cdf and quantile function for the exponential power distribution.

Finally, the second approach consist in changing the root-solving step in Algorithm 1 for the approximation of μ using the quantile function of the fitted exponential power distribution. This approach reduces the computational time, especially when dealing with larger datasets or high number of iterations in the MCMC. (se podría incluir un análisis de tiempos de computación como apéndice, para probar que efectivamente es más rápido)

3.4. Comparison criteria

Most of the goodness of fit comparison criteria for models obtained by MCMC simulation are based on the concept of deviance $\mathcal{D}(\cdot)$, which is defined as:

$$\mathcal{D}(\nu) = -2log\left(L(\nu \mid y)\right) + C,$$

where L is the likelihood function of the augmented posterior distribution, conditional on the observed data, y and C is a constant related to the saturated model, but cancels out when doing model comparisons.

The model comparison criteria that are not calculated directly from the deviance such as WAIC [31], are often multiplied by the factor -2 in order to have similar dimensions with the ones obtained with the deviance.

The widely-applicable information criterion (WAIC) is a well-known model comparison criteria used in the literature and is based on the expected log pointwise predictive posterior density [32; 33] defined for an MCMC as:

$$\widehat{\text{elppd}}_{WAIC} = \widehat{\text{lppd}} - \widehat{p}_{WAIC} \tag{22}$$

where

$$\widehat{\text{lppd}} = \sum_{i=1}^{n} log\left(\frac{1}{M}\sum_{m=1}^{M} f_Y(y_i \mid \boldsymbol{\theta}^{(m)})\right),$$
$$\widehat{p}_{WAIC} = \sum_{i=1}^{n} V_{m=1}^{M} log f_Y(y_i \mid \boldsymbol{\theta}^{(m)})$$

where $V_{m=1}^{M} x_m = \frac{1}{M-1}\sum_{m=1}^{M}(x_m - \overline{x})^2$ and M is the number of iterations. Finally, we define WAIC as -2 times equation (22) to be on the deviance scale.

Another well-known model comparison criteria in the Bayesian mainframe is the deviance information criterion (DIC) [34], defined as:

$$\text{DIC} = \overline{\mathcal{D}(\boldsymbol{\theta})} + p_D, \tag{23}$$

where $p_D = \overline{\mathcal{D}(\boldsymbol{\theta})} - \mathcal{D}(\overline{\boldsymbol{\theta}})$ is the effective number of parameters. For a MCMC, $\overline{\mathcal{D}(\boldsymbol{\theta})}$ is the mean of the deviance evaluated at each sampled value of ν and can be computed as $\overline{\mathcal{D}(\boldsymbol{\theta})} = \frac{1}{M}\sum_{m=1}^{M}\mathcal{D}(\boldsymbol{\theta}^{(m)})$, while $\mathcal{D}(\overline{\boldsymbol{\theta}})$ is the deviance evaluated at the expectation of θ.

One problem with the DIC is that is does not behave well when dealing with mixture distributions [35; 36], leading to negative number of effective parameters because of the imcomplete data problem. In this particular case, the flexible beta model is a mixture model and will be used in the simulation and application studies and will be compared against non-mixture models. In order to solve this problem, the DIC_3 was introduced [35] and its properties confirm that it can correctly estimate the number of effective parameters in a mixture model [36].

For DIC_3 the new number of parameters p_D^* is calculated as:

$$p_D^* = \overline{\mathcal{D}(\boldsymbol{\theta})} + 2log\widehat{f(\boldsymbol{y})}$$

and estimate $\widehat{f(\boldsymbol{y})} = \prod_{i=1}^{n}\widehat{f(y_i)}$ with a MCMC output we need to calculate:

$$\widehat{f(y_i)} = \frac{1}{M}\sum_{m=1}^{M}\sum_{k=1}^{K} p_k^{(m)} f_Y(y_i \mid \boldsymbol{\theta}_k^{(m)})$$

where K is the number of mixtures, $\boldsymbol{p} = [p_1, p_2, ..., p_K]^\intercal$ with $\sum p = 1$, is the weights vector and $f_Y(y \mid \boldsymbol{\theta}_k^{(m)})$ is the mixed pdf, with parameter vector of the

k-th mixture $\boldsymbol{\theta}^{(k)}$. In the particular case of the flexible beta, $f_Y(y \mid \mu_k, \phi_k)$ is the mean-variance parametrization beta pdf [3], $K = 2$ and $\phi^{(1)} = \phi^{(2)}$. More information about this distribution can be found in [4; 12]. Finally DIC_3 is calculated in the same way as DIC with the new effective number of parameters:

$$\text{DIC}_3 = \overline{\mathcal{D}(\boldsymbol{\theta})} + p_D^*. \tag{24}$$

As like every comparison criteria related to the deviance, the lower the model WAIC, DIC and DIC_3 is, the better fit it indicates, so the model with the lowest value on these criteria will be considered as the best.

4. Simulation studies

The proposed models will compete against the BIm model [7] and the FBm model (a mean reparametrized version of the Flexible Beta) [4] in 3 cases, in order to see and compare how the proposed models perform in the cases where the beta distributions fails to be a good fit. We will not use the Beta rectangular model because the flexible beta outperforms it in several cases [4; 12].

In order to make the estimations, Hamiltonian methods will be used with the software Stan [37] throughout the R package Rstan [38]. The Stan code for BIm is already implemented in [39]. The Stan code for the Flexible Beta is given in the supplementary material of [12], but we modify it in order to make our version of the FBm model. The Stan code for the Tobit-m implemented here can be found in the Appendix A, it can be easily extended from there to the other cases. One thing to account for is that the skew-normal and the root-solving functions from Stan are not as robust as the ones implemented in R packages [40; 24] and this could induce bias in the estimations because of divergences. For further reading on divergences see [41; 42]. This problem is just presented in Hamiltonian Monte Carlo, thus the Algorithm 1 will be implemented when Stan leads to divergences.

4.1. High density of values near 0 or 1

Presentar ejercicio de simulación con datasets que tengan alta densidad cerca a 1 (el caso simétrico de densidad cerca a 0 es igual y debería dar los mismos resultados).

Podemos generar datasets con media 0.8 y variando los valores de sigma para variar el peso que tendrá la densidad cuando se trunque en 1 y ver como varían los fits de las normales vs las betas. Se esperaría que mientras mayor sea sigma, mayor peso en las densidades cerca a 1 pues hay mayor dispersión y por lo tanto las distribuciones beta tendrían peor fit.

4.2. Symmetrical bell-shaped not centered in $\frac{1}{2}$

Presentar ejercicio de simulación con datasets que tengan una distribución simétrica (normal) centrada en diferentes puntos del eje Y, podría ser en 0.1, 0.3, 0.5, 0.7, 0.9, se esperaría que cuando está centrada en 0.5 el fit de las betas sea el único adecuado, en los otros casos debería empeorar pues la beta está obligada a presentar skewness y los datos son simétricos.

4.3. Right(left)-skewed on the right(left) of $\frac{1}{2}$

This simulation study challenges the performance of the Beta models (BIm and FBm) vs the proposed models (ITNm and ITSNm) under different scenarios considering distributions with skewness centered in different points. In total, 100 datasets with $n = 50$ observations each are randomly generated following what is explained below. We generate n i.i.d. observations from a uniform distribution in $(-1.5, 1.5)$ which represent a covariate x_i. Then we generate $Y_i \sim ITSNm(\gamma_i, \alpha_{0i}, \alpha_{1i}, \sigma, \lambda)$, $i = 1, ..., n$, where

$$
\begin{aligned}
logit(\gamma_i) &= \beta_{1,0} + \beta_{1,1} x_i \\
logit(\alpha_{0i}) &= \beta_{2,0} + \beta_{2,1} x_i \\
logit(\alpha_{1i}) &= \beta_{3,0} + \beta_{3,1} x_i.
\end{aligned}
$$

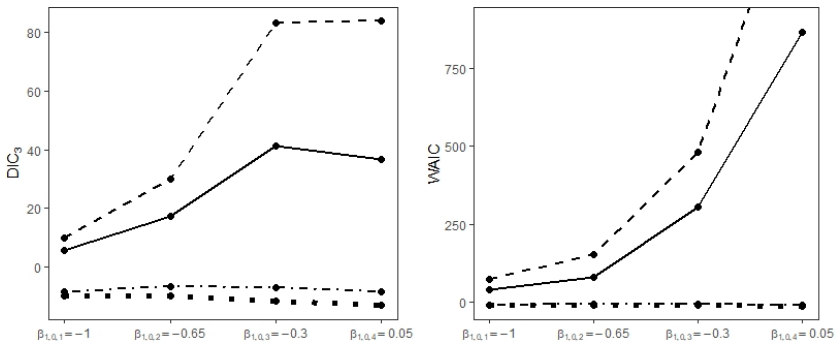
The values for each parameter are $\beta_{1,1} = -0.5, \beta_{2,0} = -1.75, \beta_{2,1} = 0.5, \beta_{3,0} = -2, \beta_{3,1} = -0.5, \sigma = 0.2$ and $\lambda = 6$. For the case of $\beta_{1,0}$ we are setting it

with 4 different values, in order to have the distribution centered in different places around the Y axis, these values are (-1, -0.65, -0.3, 0.05), this will be denoted with an extra index as $\beta_{1,0,1} = -1, \beta_{1,0,2} = -0.65, \beta_{1,0,3} = -0.3$ and $\beta_{1,0,4} = 0.05$.

The results of this simulation study are shown in Table 1 where we can see that the beta models induce bias in the estimation of the parameters related to γ and leads to having a high error the estimation of the parameters related to the inflations. Also we can see that even when considering a model with no skewness (ITNm) it induces less bias to the estimations than the beta models, due to this, we could say that it is better to consider no skewness at all than considering opposite skewness as the beta distribution is obligated to do. Regarding the fit, in Figure 6 a comparison is presented in terms of DIC_3 and WAIC, this is an obvious result for the ITSNm model being the best fit since is the model from which the data has been simulated, but it helps illustrating how the fit of the other models gets worse when translating the distribution to the right and still confirms that ITNm model fits better than the beta models when skewness that beta cannot handle are present.

Parameters	True value	BIm		FBm		ITNm	
		Bias	sd	Bias	sd	Bias	sd
$\beta_{1,0,1}$	-1	0.141	0.127	0.148	0.127	0.012	0.133
$\beta_{1,0,2}$	-0.65	0.140	0.110	0.149	0.113	-0.001	0.093
$\beta_{1,0,3}$	-0.3	0.142	0.110	0.146	0.112	-0.034	0.115
$\beta_{1,0,4}$	0.05	0.171	0.121	0.175	0.010	0.001	0.140
$\beta_{1,1}$	-0.5	0.068	0.111	0.080	0.108	0.026	0.153
$\beta_{2,0}$	-1.75	-1.171	2.612	-1.167	2.652	-0.215	0.654
$\beta_{2,1}$	0.5	0.257	1.978	0.313	2.014	-0.155	0.656
$\beta_{3,0}$	-2	-2.17	2.908	-2.153	2.894	-0.132	0.598
$\beta_{3,1}$	-0.5	-0.705	1.866	-0.685	1.851	0.218	0.572

Table 1: Parameter recovery results for simulation study on skewness



(a) DIC_3 results

(b) WAIC results

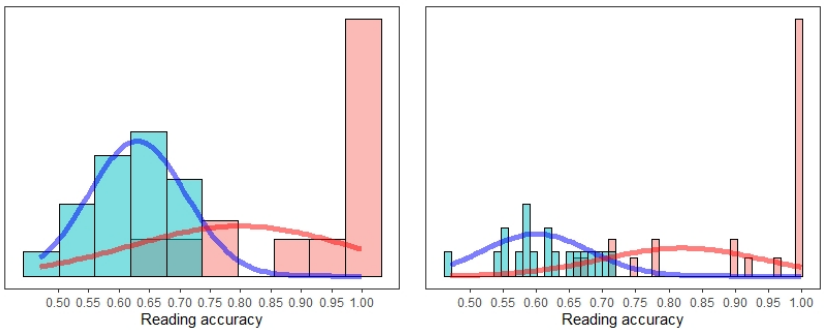
Figure 6: The x axis represents the value of $\beta_{1,0}$ took on the generation of the datasets, and y axis represents the mean fit criteria. Figure (a) shows the results for fit comparing DIC_3, in order of goodness of fit: the squared-dot line is the ITSNm, the dot-dashed line is the ITNm (really close to ITSNm but a little worse), the solid line is the BIm and the dashed line is the FBm. Figure (b) shows the results for fit comparing WAIC, in order of goodness of fit: the squared-dot line is the ITSNm, the dot-dashed line is the ITNm (really close to ITSNm but a little worse), the solid line is the BIm and the dashed line is the FBm, the plot is trimmed from the upper side due to a high value on WAIC.

5. Application to real data

5.1. Reading accuracy data

The following dataset consist of 44 children, 19 of whom have been diagnosed with dyslexia [1]. The response variable, y, quantifies the reading accuracy performance as a proportion $(0, 1]$, as plotted in Figure 7, while a quantitative covariate, x, measures the standardized nonverbal IQ of the children. This dataset has been used in several papers [6; 43; 44; 4; 12], which highlights the interest on correctly modelling this problem. Despite this, just [12] models it using inflation, but does not models the marginal mean directly through the covariates IQ and dyslexic status.

The proposed analysis consists in investigating the fit of the proposed models (Tobit-m, ITNm and ITSNm) on the Reading accuracy dataset. Here we assume a functional relationship between intelligence (quantified through the standarized IQ named x), dyslexic status (a dichotomous covariate d reflecting



(a) Reading accuracy in 10 bins (b) Reading accuracy in 25 bins

Figure 7: The same responses are plotted considering the different status, where blue represents dyslexic children and red represent non-dyslexic children. Also, different number of bins are considered, in (a) 10 bins and in (b) 25 bins. Normal curves are fitted to the dyslexic and non-dyslexic distributions.

being dyslexic (1) or not (0)) and the reading accuracy y as response variable. For this purpose we define the following regression model that includes regression functions for the marginal mean, the probability of 1 and the standard deviation:

$$\begin{aligned} logit(\gamma_i) &= \beta_{1,0} + \beta_{1,1} d_i \\ logit(\alpha_{1i}) &= \beta_{2,0} + \beta_{2,1} d_i \\ logit(\sigma_i) &= \beta_{3,0} + \beta_{3,1} d_i + \beta_{3,2} x_i + \beta_{3,3} x_i d_i. \end{aligned} \tag{25}$$

For comparison purposes, we estimate the BIm [7] and FBm [4] modelling $logit(\phi_i) = \beta_{3,0} + \beta_{3,1} d_i + \beta_{3,2} x_i + \beta_{3,3} x_i d_i$ instead of σ. The effects of covariate x and the interaction between covariates x and d where removed for modelling γ and α_1 due to lack of significance in all competing models.

WAIC, DIC and DIC_3 results for each model and parameter estimation are in Table 2. DIC results have been left in the table but is worth emphasizing that should not be considered in the cases of mixtures (FBm) as discussed in 3.4.

Results indicate a better fit for the ITNm model according to WAIC and DIC_3 comparing all models, and according to DIC (excluding FBm). The IT-

SNm model is not included because all the parameters related to the skewness parameter λ resulted non significant and therefore did not add value to the fit. It should be also noted that regardless not being the best fit, Tobit-m performs well, giving close competition to the other models without modelling inflation directly, and the estimation of the mean parameters is really close to the ones achieved by the best fit (ITNm). The main difference with the Tobit-m can be noticed in the estimation of the standard deviation, where in order to model the inflation in 1, Tobit-m has to "sacrifice" its precision by making $\beta_{3,0}$ larger so it accumulates more density in $Y = 1$.

Algorithm 1: Root-solving Metropolis-Hastings algorithm (ITNm)

Initialize $\boldsymbol{\theta}^{(0)} = (\gamma^{(0)}, \alpha_a^{(0)}, \alpha_b^{(0)}, \sigma^{(0)})$

Compute:

$\delta_a^{(0)} = \alpha_a^{(0)} \cdot (1 - \gamma^{(0)})$

$\delta_b^{(0)} = \alpha_b^{(0)} \cdot \gamma^{(0)}$

With $\psi = \frac{a-\mu}{\sigma}$ and $\psi^\dagger = \frac{b-\mu}{\sigma}$,

$$f(\mu) = \delta_a a + \delta_b b + (1 - \delta_a - \delta_b) \left[\mu - \sigma \frac{\phi(\psi^\dagger) - \phi(\psi)}{\Phi(\psi^\dagger) - \Phi(\psi)} \right] - \gamma$$

Solve equation:

$$f(\mu^{(0)} | \gamma^{(0)}, \delta_a^{(0)}, \delta_b^{(0)}, \sigma^{(0)}) = 0 \iff \mu^{(0)} = f^{-1}(0 | \gamma^{(0)}, \delta_a^{(0)}, \delta_b^{(0)}, \sigma^{(0)})$$

for *i=1,2,...* **do**

Sample candidate: $\boldsymbol{\theta}^{cand} \sim q(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)})$

Compute:

$\delta_a^{cand} = \alpha_a^{cand} \cdot (1 - \gamma^{cand})$

$\delta_b^{cand} = \alpha_b^{cand} \cdot \gamma^{cand}$

$\mu^{cand} = f^{-1}(0 \mid \gamma^{cand}, \delta_a^{cand}, \delta_b^{cand}, \sigma^{cand})$

Calculate Acceptance Probability:

$$\alpha = \min \left\{ 1, \frac{L(\boldsymbol{y} | \mu^{cand}, \delta_a^{cand}, \delta_b^{cand}, \sigma^{cand})}{L(\boldsymbol{y} | \mu^{(i-1)}, \delta_a^{(i-1)}, \delta_b^{(i-1)}, \sigma^{(i-1)})} \frac{P(\boldsymbol{\theta}^{cand})}{P(\boldsymbol{\theta}^{(i-1)})} \frac{q(\boldsymbol{\theta}^{(i-1)} | \boldsymbol{\theta}^{cand})}{q(\boldsymbol{\theta}^{cand} | \boldsymbol{\theta}^{(i-1)})} \right\}$$

Sample uniform: $u \sim$ Uniform(0,1)

if $u < \alpha$ **then**

Accept the proposal: $\boldsymbol{\theta}^{(i)} \leftarrow \boldsymbol{\theta}^{cand}$

else

Reject the proposal: $\boldsymbol{\theta}^{(i)} \leftarrow \boldsymbol{\theta}^{(i-1)}$

end

end

Parameters	BIm		FBm		Tobit-m		ITNm	
	mean	CI	mean	CI	mean	CI	mean	CI
$\beta_{1,0}$ (intercept)	2.158	(1.631;2.761)	2.108	(1.600;2.716)	2.230	(1.574;2.950)	2.259	(1.740;2.853)
$\beta_{1,1}$ (dyslexic)	−1.658	(-2.284;-1.111)	−1.598	(-2.249;-1.031)	−1.770	(-2.492;-1.093)	−1.801	(-2.404;-1.245)
$\beta_{2,0}$ (intercept)	0.222	(-0.555;1.050)	0.242	(-0.500;1.005)			0.300	(-0.472;1.090)
$\beta_{2,1}$ (dyslexic)	−3.812	(-6.862;-1.562)	−9.281	(-37.150;-1.359)			−7.050	(-10.931;-3.430)
$\beta_{3,0}$ (intercept)	2.445	(1.430;3.235)	2.964	(1.573;4.727)	−1.486	(-1.899;-0.932)	−1.826	(-2.534;-0.524)
$\beta_{3,1}$ (dyslexic)	1.814	(0.492;3.123)	1.459	(-0.709;3.443)	−1.303	(-2.005;-0.640)	−1.008	(-2.413;-0.097)
$\beta_{3,2}$ (IQ)	−1.217	(-2.148;-0.188)	−1.775	(-3.928;-0.312)	0.273	(-0.342;0.859)	0.728	(-0.215;1.496)
$\beta_{3,3}$ (IQ × dyslexic)	1.635	(0.120;3.026)	2.188	(0.276;4.597)	−0.3857	(-1.1045;0.3222)	−0.919	(-1.834;0.127)
WAIC	−27.545		−26.772		−26.925		**−28.225**	
p_{waic}	5.695		7.460		4.510		5.216	
DIC	−26.944		~~-35.921~~		−26.571		**−28.263**	
p_d	7.166		~~-0.876~~		5.285		5.784	
DIC_3	−29.286		−28.397		−27.769		**−29.437**	
p_d^*	4.824		6.648		4.088		4.609	

Table 2: Results for modelling with covariates significant in at least one model

5.2. Lung function data

This second dataset was obtained by [45] and captures information of 3146 children about their spirometric lung function (slf) as the response variable, which is defined as $slf = FEV_1/FVC$, where the FEV_1 is the forced expiratory volume in 1 second and FVC is the forced vital capacity. This response variable is an established index for diagnosing airway obstruction [46]. As covariate the height of the children is recorded. This dataset is also used in statistical literature in [47; 48].

The proposed regression:

$$\begin{aligned} logit(\gamma_i) &= \beta_{1,0} + \beta_{1,1} log(height) \\ logit(\alpha_{1i}) &= \beta_{2,0} + \beta_{2,1} log(height) \\ logit(\sigma_i) &= \beta_{3,0} + \beta_{3,1} log(height). \end{aligned} \tag{26}$$

For comparison purposes, we estimate the BIm [7] and FBm [4] modelling $logit(\phi_i) = \beta_{3,0} + \beta_{3,1}\, log(height)$ instead of σ.

Parameters	BIm		FBm		Tobit-m	
	mean	CI	mean	CI	mean	CI
$\beta_{1,0}$ (intercept)	2.148	(2.121;2.174)	2.140	(2.112;2.168)	2.128	(2.102;2.152)
$\beta_{1,1}$ (log(height))	−2.487	(-2.614;-2.360)	−2.481	(-2.607;-2.352)	−2.410	(-2.522;-2.291)
$\beta_{2,0}$ (intercept)	−3.026	(-3.268;-2.789)	−3.092	(-3.355;-2.839)		
$\beta_{2,1}$ (log(height))	−8.236	(-9.326;-7.190)	−8.547	(-9.785;-7.327)		
$\beta_{3,0}$ (intercept)	3.029	(2.978;3.082)	3.199	(3.129;3.265)	−2.622	(-2.649;-2.595)
$\beta_{3,1}$ (log(height))	0.198	(-0.057;0.464)	0.800	(0.522;1.099)	0.042	(-0.087;0.169)
WAIC	−6147.452		−6244.598		−6193.045	
p_{waic}	6.649		8.864		5.400	
DIC	−6148.471		−6245.913		−6194.407	
p_d	5.728		7.690		4.053	
DIC_3	−6147.647		−6244.88		−6193.075	
p_d^*	6.552		8.723		5.385	

Table 3: Results for modelling the Lung function data

6. Discussion

We presented three normal-like models that... showing better fit than the ones in the literature for modelling bounded inflated responses.

Simulation studies and applications where conducted, leading to a better fit for out proposed models.

Acknowledgments

References

[1] Kristen Pammer and Alison Kevan. The contribution of visual sensitivity, phonological processing, and nonverbal iq to children's reading. *Scientific Studies of Reading*, 11:33–53, 01 2007. doi: 10.1080/10888430709336633.

[2] NINDS rt PA Stroke Study Group. Tissue plasminogen activator for acute ischemic stroke. *New England Journal of Medicine*, 333(24):1581–1588, 1995. doi: 10.1056/NEJM199512143332401. URL https://doi.org/10.1056/NEJM199512143332401. PMID: 7477192.

[3] Silvia Ferrari and Francisco Cribari-Neto. Beta regression for modelling rates and proportions. *Journal of Applied Statistics*, 31(7):799–815, 2004. doi: 10.1080/0266476042000214501. URL https://doi.org/10.1080/0266476042000214501.

[4] Sonia Migliorati, Agnese Maria Di Brisco, and Andrea Ongaro. A new regression model for bounded responses. *Bayesian Anal.*, 13(3):845–872, 09 2018. doi: 10.1214/17-BA1079. URL https://doi.org/10.1214/17-BA1079.

[5] Cristian L. Bayes, Jorge L. Bazán, and Catalina García. A new robust regression model for proportions. *Bayesian Anal.*, 7(4):841–866, 12 2012. doi: 10.1214/12-BA728. URL https://doi.org/10.1214/12-BA728.

[6] Michael Smithson and Jay Verkuilen. A better lemon squeezer? maximum-likelihood regression with beta-distributed dependent variables. *Psychological methods*, 11:54–71, 04 2006. doi: 10.1037/1082-989X.11.1.54.

[7] Cristian L. Bayes and Luis Valdivieso. A beta inflated mean regression model for fractional response variables. *Journal of Applied Statistics*, 43 (10):1814–1830, 2016. doi: 10.1080/02664763.2015.1120711. URL https://doi.org/10.1080/02664763.2015.1120711.

[8] Raydonal Ospina and Silvia LP Ferrari. Inflated beta distributions. *Statistical papers*, 51(1):111, 2010.

[9] Raydonal Ospina and Silvia L.P. Ferrari. A general class of zero-or-one inflated beta regression models. *Computational Statistics & Data Analysis*, 56 (6):1609–1623, 2012. URL https://EconPapers.repec.org/RePEc:eee:csdana:v:56:y:2012:i:6:p:1609-1623.

[10] BEYGI A MOGHIM, MR Eshraghian, KAZEM MOHAMMAD, JELIANI KA NOURI, and M HOSSEINI. Determinants number of cigarette smoked with iranian adolescents: A multilevel zero inflated poisson regression model. 2009.

[11] David Todem, KyungMann Kim, and Wei-Wen Hsu. Marginal mean models for zero-inflated count data. *Biometrics*, 72(3):986–994, 2016. doi: 10.1111/biom.12492. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/biom.12492.

[12] Agnese Maria Di Brisco and Sonia Migliorati. A new mixed-effects mixture model for constrained longitudinal data. *Statistics in medicine*, 39(2):129–145, 2020.

[13] Eugene Hahn. Mixture densities for project management activity times: A robust approach to pert. *European Journal of Operational Research*, 188: 450–459, 02 2008. doi: 10.1016/j.ejor.2007.04.032.

[14] Jue Wang and Sheng Luo. Augmented beta rectangular regression models: A bayesian perspective. *Biometrical journal. Biometrische Zeitschrift*, 58(1):206—221, January 2016. ISSN 0323-3847. doi: 10.1002/bimj.201400232. URL https://europepmc.org/articles/PMC5064841.

[15] James Tobin. Estimation of relationships for limited dependent variables. Cowles Foundation Discussion Papers 3R, Cowles Foundation for Research in Economics, Yale University, 1956. URL https://EconPapers.repec.org/RePEc:cwl:cwldpp:3r.

[16] John McDonald and Robert Moffitt. The uses of tobit analysis. *The Review of Economics and Statistics*, 62(2):318–21, 1980. URL https://EconPapers.repec.org/RePEc:tpr:restat:v:62:y:1980:i:2:p:318-21.

[17] Michael R Frone, M Lynne Cooper, and Marcia Russell. Stressful life events, gender, and substance use: An application of tobit regression. *Psychology of Addictive Behaviors*, 8(2):59, 1994.

[18] Ming-Te Wang, Robert L Selman, Thomas J Dishion, and Elizabeth A Stormshak. A tobit regression analysis of the covariation between middle school students' perceived school climate and behavioral problems. *Journal of Research on adolescence*, 20(2):274–286, 2010.

[19] John Burkardt. The truncated normal distribution. 2014.

[20] A. Azzalini. A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics*, 12(2):171–178, 1985. ISSN 03036898, 14679469. URL http://www.jstor.org/stable/4615982.

[21] Samir K. Ashour and Mahmood A. Abdel-hameed. Approximate skew normal distribution. *Journal of Advanced Research*, 1(4):341 – 350, 2010. ISSN 2090-1232. doi: https://doi.org/10.1016/j.jare.2010.06.004. URL http://www.sciencedirect.com/science/article/pii/S209012321000069X.

[22] Cedric Flecher, Denis Allard, and Philippe Naveau. Truncated skew-normal distributions: Moments, estimation by weighted moments and application to climatic data. *Metron - International Journal of Statistics*, LXVIII: 331–345, 12 2010. doi: 10.1007/BF03263543.

[23] Cristian Luis Bayes and Márcia D'Elia Branco. Bayesian inference for the skewness parameter of the scalar skew-normal distribution. *Brazilian Journal of Probability and Statistics*, 21(2):141–163, 2007. ISSN 01030752, 23176199. URL http://www.jstor.org/stable/43601095.

[24] Karline Soetaert. Package rootsolve: roots, gradients and steady-states in r. *URL https://cran. r-project. org/web/packages/rootSolve/vignettes/-rootSolve. pdf*, 2014.

[25] Omar M Eidous and Rima Abu-Shareefa. New approximations for standard normal distribution function. *Communications in Statistics-Theory and Methods*, 49(6):1357–1374, 2020.

[26] Amit Choudhury. A simple approximation to the area under standard normal curve. *Mathematics and Statistics*, 2(3):147–149, 2014.

[27] Werner Hürlimann. Tail approximation of the skew-normal by the skew-normal laplace: Application to owen's t function and the bivariate normal distribution. *Journal of Statistical and Econometric Methods*, 2(1):1–12, 2013.

[28] Donald Bruce Owen. A table of normal integrals: A table. *Communications in Statistics-Simulation and Computation*, 9(4):389–419, 1980.

[29] AA Olosunde. On exponential power distribution and poultry feeds data: a case study. *Journal of The Iranian Statistical Society*, 12(2):253–270, 2013.

[30] Mikis Stasinopoulos, Bob Rigby, Calliope Akantziliotou, Gillian Heller, Raydonal Ospina, and Maintainer Mikis Stasinopoulos. Package 'gamlss. dist', 2020.

[31] Sumio Watanabe. Asymptotic equivalence of bayes cross validation and widely applicable information criterion in singular learning theory. *Journal of Machine Learning Research*, 11(Dec):3571–3594, 2010.

[32] Andrew Gelman, John B Carlin, Hal S Stern, David B Dunson, Aki Vehtari, and Donald B Rubin. *Bayesian data analysis*. CRC press, 2013.

[33] Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27(5):1413–1432, 2017.

[34] David J Spiegelhalter, Nicola G Best, Bradley P Carlin, and Angelika Van Der Linde. Bayesian measures of model complexity and fit. *Journal of the royal statistical society: Series b (statistical methodology)*, 64(4):583–639, 2002.

[35] Gilles Celeux, Florence Forbes, Christian P Robert, D Mike Titterington, et al. Deviance information criteria for missing data models. *Bayesian analysis*, 1(4):651–673, 2006.

[36] Chanmin Kim. Deviance information criteria for mixtures of distributions. *Communications in Statistics-Simulation and Computation*, pages 1–14, 2019.

[37] Bob Carpenter, Andrew Gelman, Matthew Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software, Articles*, 76(1):1–32, 2017. ISSN 1548-7660. doi: 10.18637/jss.v076.i01. URL https://www.jstatsoft.org/v076/i01.

[38] Stan Development Team. RStan: the R interface to Stan, 2020. URL http://mc-stan.org/. R package version 2.19.3.

[39] Renzo Fernández, Cristian L Bayes, and Luis Valdivieso. A beta-inflated mean regression model with mixed effects for fractional response variables. *Journal of Statistical Computation and Simulation*, 88(10):1936–1957, 2018.

[40] Adelchi Azzalini, Maintainer Adelchi Azzalini, and T Owen. The sn package. *R Reference Guide available from www. r-project. org*, 2004.

[41] Michael Betancourt. Diagnosing suboptimal cotangent disintegrations in hamiltonian monte carlo. 2016.

[42] Michael Betancourt. Diagnosing biased inference with divergences @ONLINE, January 2017. URL https://mc-stan.org/users/documentation/case-studies/divergences_and_bias.html.

[43] Patrícia L. Espinheira, Silvia L.P. Ferrari, and Francisco Cribari-Neto. On beta regression residuals. *Journal of Applied Statistics*, 35(4):407–419, 2008. doi: 10.1080/02664760701834931. URL https://doi.org/10.1080/02664760701834931.

[44] Francisco Cribari-Neto and Achim Zeileis. Beta regression in r. *Journal of Statistical Software, Articles*, 34(2):1–24, 2010. ISSN 1548-7660. doi: 10.18637/jss.v034.i02. URL https://www.jstatsoft.org/v034/i02.

[45] Sanja Stanojevic, Angie Wade, Tim J Cole, Sooky Lum, Adnan Custovic, Mike Silverman, Graham L Hall, Liam Welsh, Jane Kirkby, Wenche Nystad, et al. Spirometry centile charts for young caucasian children: the asthma uk collaborative initiative. *American journal of respiratory and critical care medicine*, 180(6):547, 2009.

[46] PH Quanjer, Sanja Stanojevic, Janet Stocks, GL Hall, KVV Prasad, TJ Cole, M Rosenthal, R Perez-Padilla, JL Hankinson, Emanuela Falaschetti, et al. Changes in the fev1/fvc ratio during childhood and adolescence: an intercontinental study. *European Respiratory Journal*, 36 (6):1391–1399, 2010.

[47] Abu Hossain, Robert Rigby, Mikis Stasinopoulos, and Marco Enea. Centile estimation for a proportion response variable. *Statistics in medicine*, 35(6): 895–904, 2016.

[48] Robert A Rigby, Mikis D Stasinopoulos, Gillian Z Heller, and Fernanda De Bastiani. *Distributions for modeling location, scale, and shape: Using GAMLSS in R*. CRC press, 2019.

Appendices

A. Stan Code

```
functions {
  real tobit_lpdf(real y, real mu, real sigma, real a, real b) {
    real prob;
    real alpha;
    real beta;

    alpha = (a-mu)/sigma;
    beta = (b-mu)/sigma;

    if(y <= a){
      prob = log(Phi_approx(alpha));
    } else if(y >= b){
      prob = log(1-Phi_approx(beta));
    } else {
      prob = std_normal_lpdf(((y-mu)/sigma)) - log(sigma);
    }
    return(prob);
  }

  vector system(vector mu, vector theta, real[] x_r, int[] x_i){
    vector[1] z; //required vector in system solving
```

```
real a;
real b;

real sigma;
real gamma;

real alpha;
real beta;

real X; //temporary variable
real W; //temporary variable

a = x_r[1];
b = x_r[2];

sigma = theta[1];
gamma = theta[2];

alpha = (a-mu[1])/sigma;
beta = (b-mu[1])/sigma;

X = exp(log_diff_exp(log(Phi_approx(beta)),
        log(Phi_approx(alpha))));
W = exp(std_normal_lpdf(beta))*
        (1 - exp(std_normal_lpdf(alpha)
                - std_normal_lpdf(beta)));

z[1] = Phi_approx(alpha)*a +
        (1-Phi_approx(beta))*b +
        mu[1]*X - sigma*W - gamma;
```

```
    return(z);
  }
}

data {
  int n;
  int n_cov_gamma; //number of covariates to model gamma
  int n_cov_sigma; //number of covariates to model sigma
  int n_pars;
  vector[n] y;
  matrix[n, n_cov_gamma] x_gamma; //X matrix for gamma
  matrix[n, n_cov_sigma] x_sigma; //X matrix for sigma
  real a;
  real b;
}

transformed data{
  vector[1] mu_guess;
  real x_r[2];
  int x_i[0]; //required for system solving

  mu_guess[1] = (a+b)/2; //initial point for solving eq
  x_r = {a, b};
}

parameters {
  vector[n_cov_gamma] beta_gamma;
  vector[n_cov_sigma] beta_sigma;
}
```

```
transformed parameters {
  vector[n] linpred_gamma;
  vector[n] gamma;
  vector[n] linpred_sigma;
  vector[n] sigma;
  vector[1] mu[n]; //mu must be declared as vector for system solving
  vector[n_pars] theta;

  linpred_gamma = x_gamma * beta_gamma;
  linpred_sigma = x_sigma * beta_sigma;

  for(i in 1:n){
    gamma[i] = inv_logit(linpred_gamma[i]);
    sigma[i] = exp(linpred_sigma[i]);
    theta[1] = sigma[i];
    theta[2] = gamma[i];
    mu[i] = algebra_solver(system, mu_guess, theta, x_r, x_i,
                           1e-10, 1e-5, 1e5);
  }
}

model {
  //Priors
  for(i in 1:n_cov_gamma) beta_gamma[i] ~ normal(0, 5);
  for(i in 1:n_cov_sigma) beta_sigma[i] ~ normal(0, 5);

  for(i in 1:n) y[i] ~ tobit(mu[i, 1], sigma[i], a, b);
}
```